How Many Clusters?

Evaluating K in K-Means Using the Elbow Method and Silhouette Score

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GitHub: https://github.com/Easywiz/K-Means-Elbow-vs-Silhouette

Abstract. K-Means clustering requires the number of clusters K to be specified in advance — yet in most real-world scenarios, the true number of clusters is unknown. This tutorial introduces and compares two principled methods for evaluating K: the **Elbow Method**, based on within-cluster variance (inertia), and the **Silhouette Score**, which balances cluster cohesion with separation. We demonstrate both on a synthetic dataset (where ground truth is known) and on the UCI Wine dataset, revealing when they agree, when they disagree, and which to trust.

1. Introduction — The Problem of Choosing K

K-Means is one of the most widely used unsupervised learning algorithms. Given a dataset of N points, it partitions them into K clusters by iteratively assigning each point to the nearest centroid and updating centroids to the mean of their assigned points. The objective is to minimize **inertia** — the total within-cluster sum of squared distances.

The catch is fundamental: **K must be chosen before the algorithm runs.** A poor choice corrupts all downstream analysis. Choose K=2 on data with five natural groups and you will merge distinct populations. Choose K=10 on the same data and you will fragment cohesive clusters into meaningless fragments.

This tutorial teaches two methods for selecting K systematically, explains the intuition behind each, and shows — critically — how to interpret them when they give conflicting answers.

2. How K-Means Works — A Brief Recap

The K-Means algorithm proceeds as follows:

- **Initialize:** Place K centroids, ideally using K-Means++ (Arthur & Vassilvitskii, 2007), which seeds centroids far apart to avoid poor local minima.
- **Assign:** Assign each data point to the nearest centroid using Euclidean distance.
- **Update:** Recompute each centroid as the mean of all points assigned to it.
- **Repeat:** Iterate steps 2–3 until centroid positions no longer change (convergence).

The algorithm minimizes inertia: $\sum_{i=1}^{N} \min_k |x_i - \mu_k|^2$ Critically, inertia *always* decreases as K increases — reaching zero when K equals N. This means we cannot simply minimize inertia to find the best K; we need smarter criteria.

Important: Always standardize your features before running K-Means. Because the algorithm relies on Euclidean distance, features with large scales (e.g., proline content: 0–1680) will dominate over features with small scales (e.g., hue: 0.5–1.7), regardless of their actual importance.

3. Dataset 1 — Synthetic Data (Known Ground Truth)

Before applying evaluation methods to real data, we test them on a synthetic dataset where we know the true answer. We generate 400 points from 4 Gaussian blobs using scikit-learn's *make_blobs*. The algorithm sees unlabeled points (Figure 1, left); we use the ground truth (right) only to verify our methods.

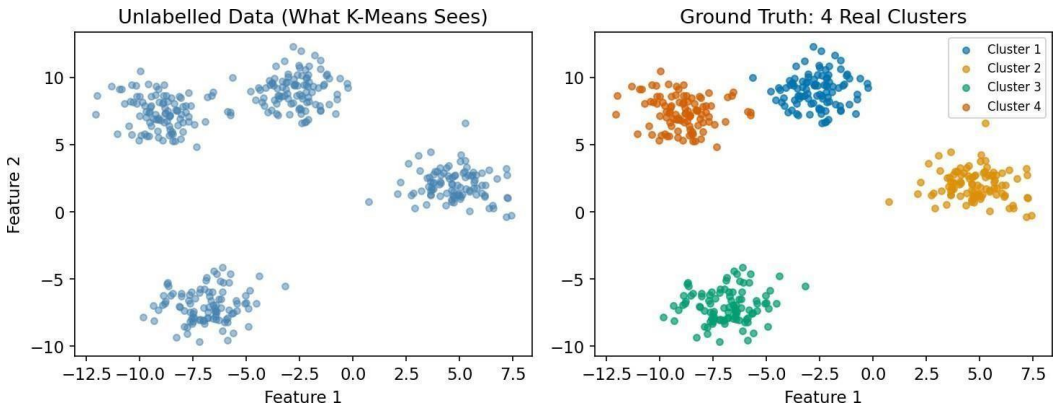

Figure 1: Left — raw unlabeled data as K-Means sees it. Right — ground truth showing 4 real clusters (colours added for verification only). Colourblind-friendly palette used throughout.

4. The Elbow Method

4.1 Intuition

We fit K-Means for K = 1 to 12 and plot inertia against K. At low K, adding an extra cluster dramatically reduces inertia because we are splitting large, poorly-fitted clusters. Beyond the true K, additional clusters offer only marginal improvement — the curve 'elbows' and flattens. The optimal K is at this bend.

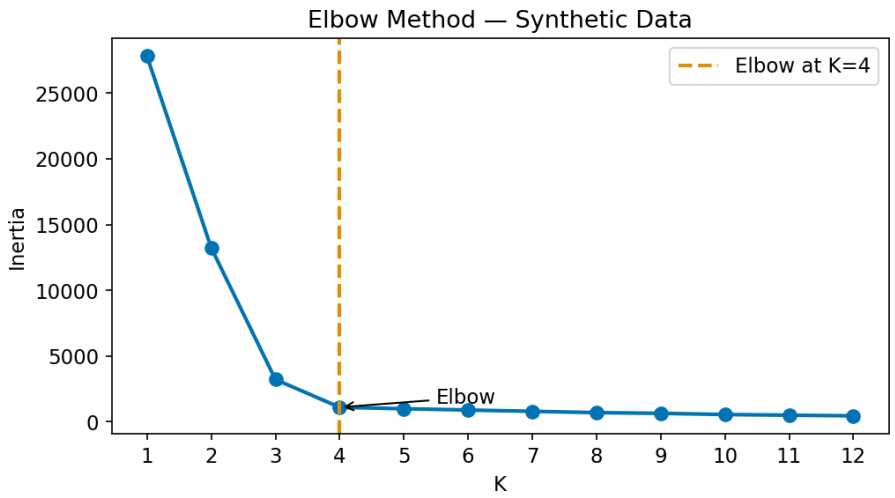

Figure 2: Elbow curve for the synthetic dataset. The dashed line marks K=4, where inertia stops decreasing sharply — correctly identifying the true number of clusters.

4.2 Limitation

The elbow is not always obvious. When clusters overlap or vary in density, the curve may be smooth with no clear bend. In these cases, the Elbow Method is unreliable and the Silhouette Score should be preferred.

5. The Silhouette Score

5.1 Intuition

The Silhouette Score (Rousseeuw, 1987) evaluates clustering quality by measuring how well each point fits its assigned cluster compared to neighboring clusters. For each point i, two quantities are computed:

- **a(i)** — mean distance to all other points in the same cluster (*cohesion*). Lower is better.
- **b(i)** — mean distance to all points in the nearest *other* cluster (*separation*). Higher is better.

The silhouette for point i is: $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$. Scores range from –1 to +1.

Score Range	Interpretation
Close to +1	Point is well-matched to its cluster
Close to 0	Point lies on the boundary between two clusters
Close to –1	Point may have been assigned to the wrong cluster

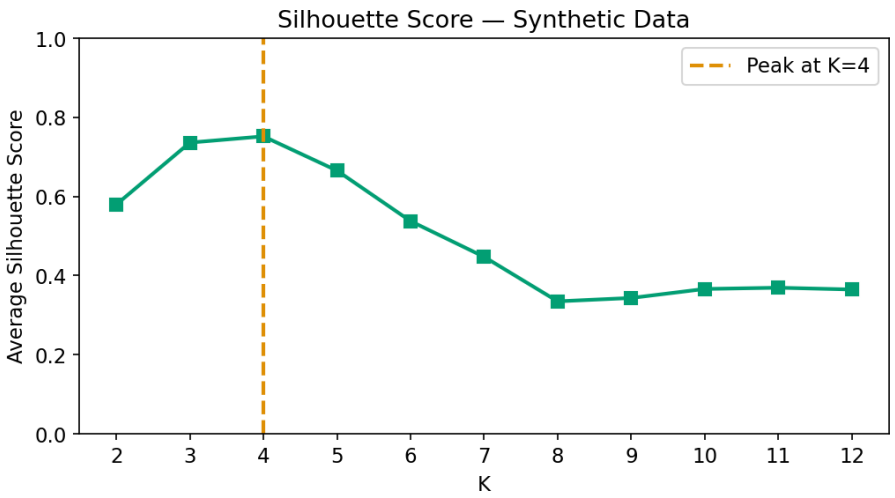

Figure 3: Average silhouette score vs K for the synthetic dataset. Score peaks clearly at K=4, confirming the true number of clusters. Unlike inertia, higher score = better clustering.

5.2 Silhouette Diagrams — Per-Cluster Breakdown

Rather than just using the average score, silhouette diagrams visualize every individual point. Each cluster is shown as a horizontal band; wider bands indicate more points. Bars extending past the average line (red dashes) are well-classified. Negative bars signal misclassified points.

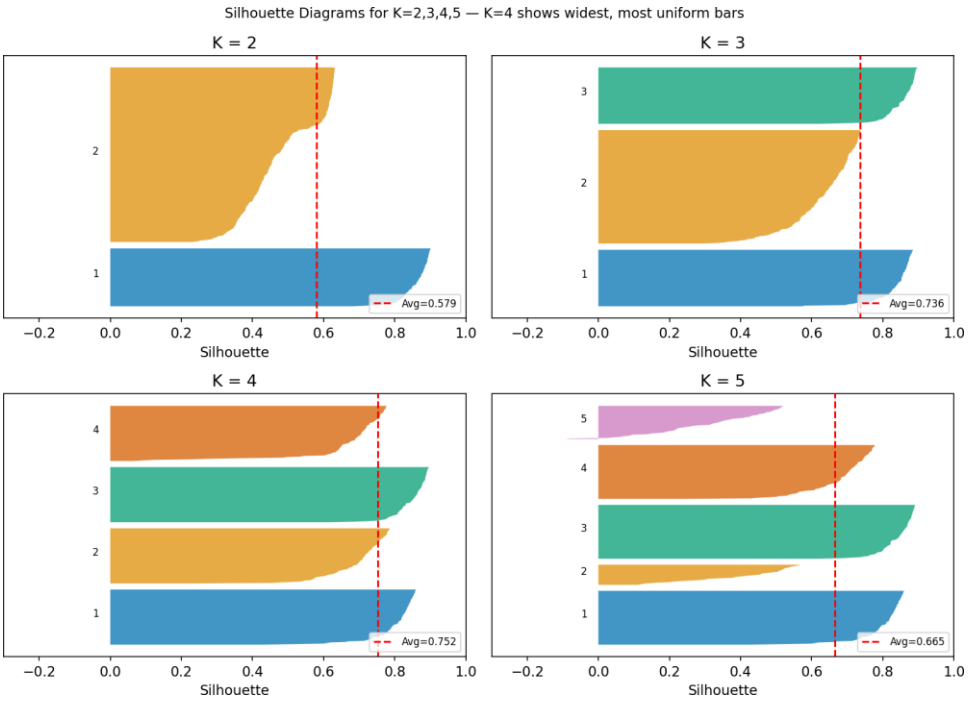

Figure 4: Silhouette diagrams for K=2, 3, 4, 5 on synthetic data. K=4 (bottom-left) shows the widest and most uniform bars with no negative values — the clearest sign of correct clustering.

Notice that at K=2 and K=3, some clusters are thin and some bars dip below zero. At K=5, one cluster is very small, suggesting it has split a natural group unnecessarily. Only K=4 shows four balanced, wide bands — all well above the average.

6. Dataset 2 — Real World: UCI Wine Data

We now apply both methods to the UCI Wine dataset: 178 wine samples described by 13 chemical measurements (alcohol, malic acid, ash, etc.). Unlike the synthetic case, we do not reveal the true cluster count — this is a genuine evaluation challenge.

Before clustering, all 13 features are standardized using *StandardScaler*. Without this step, proline (range: 278–1680) would dominate the distance calculations and produce meaningless clusters.

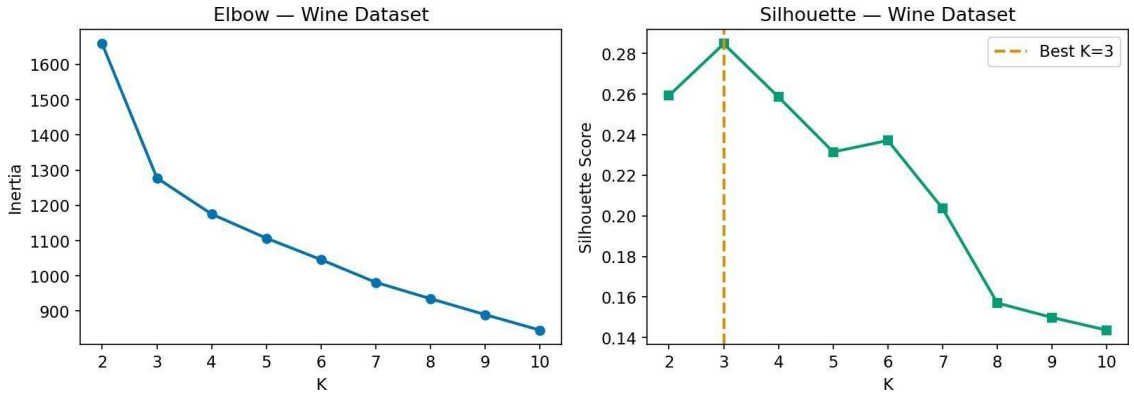

Figure 5: Elbow method (left) and silhouette score (right) for the Wine dataset. The elbow suggests K≈3; the silhouette score peaks clearly at K=3. Here, both methods agree.

Both methods converge on **K=3** for this dataset — a satisfying result, since the Wine dataset was originally collected from three cultivars (grape varieties). This is a case where the methods agree and domain knowledge confirms the answer.

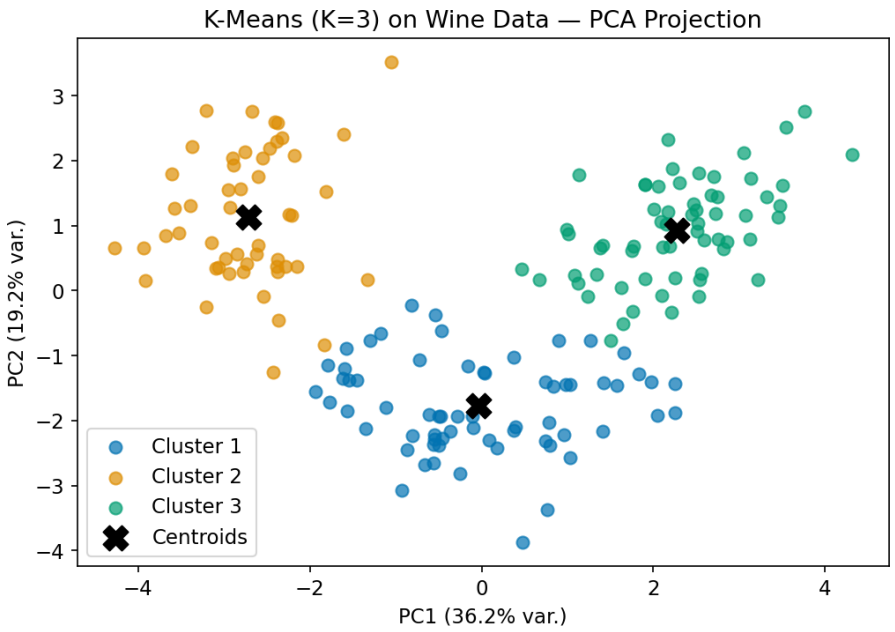

Figure 6: Final K=3 clustering on the Wine dataset, visualized by projecting onto the first two principal components (55.4% variance explained). Black X markers indicate cluster centroids.

7. When the Methods Disagree

When clusters are well-separated, the Elbow Method and Silhouette Score typically agree. On overlapping or irregular data they may diverge. As a practical guide:

Situation	Recommendation
Methods agree	Use the agreed K with confidence
Methods disagree	Try both K values; use domain knowledge to decide
No clear elbow	Prefer Silhouette Score
Large dataset (slow silhouette)	Elbow first; validate silhouette on a sample
Clusters non-spherical or unequal size	Consider DBSCAN or Gaussian Mixture Models

Our Jupyter notebook includes a demonstration with overlapping blobs where the elbow curve shows no clear bend, yet the silhouette score still correctly identifies the optimal K. This illustrates why silhouette diagrams — not just average scores — are valuable: they reveal not only the best K but also how confidently the data supports that choice.

8. Conclusion

Choosing K is not guesswork. The Elbow Method provides a fast, intuitive first pass by looking for diminishing returns in inertia. The Silhouette Score provides a richer, more reliable evaluation by measuring both within-cluster cohesion and between-cluster separation.

The five key takeaways from this tutorial:

- Inertia always decreases — the Elbow Method looks for the bend, not the minimum.
- Silhouette Score ranges from −1 to +1; higher is always better.
- Use silhouette *diagrams*, not just the average, to evaluate individual clusters.
- Always standardize features before running K-Means.
- Use both methods together and validate with domain knowledge.

Try it yourself: The full Jupyter notebook for this tutorial — including all code, figures, and intermediate steps — is available at: **https://github.com/Easywiz/K-Means-Elbow-vs-Silhouette** The repository includes a README, MIT license, and instructions for reproducing all results

Accessibility Considerations

This tutorial has been designed with accessibility in mind throughout. All figures use a colourblind-friendly palette (seaborn's colorblind preset), ensuring that colour is never the sole means of conveying information — cluster distinctions remain visible across the most common forms of colour vision deficiency. Every figure in the accompanying Jupyter notebook includes printed alt-text describing its content, making the code outputs accessible to screen reader users. The PDF itself uses a clean, high-contrast layout with clearly labelled axes, descriptive captions for all figures, and structured section headings to support navigation. Tables are used in place of prose lists where comparisons are made, improving readability for all audiences.

References

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UCI Machine Learning Repository *Wine Dataset*. Available at: https://archive.ics.uci.edu/ml/datasets/wine

License: MIT | GitHub: https://github.com/Easywiz/K-Means-Elbow-vs-Silhouette | Colourblind-friendly figures (colorblind palette, seaborn) | All figures include alt-text in the accompanying notebook.